# ROAD DAMAGE AND ACCIDENT RISK ANALYTICS DASHBOARD USING BUSINESS INTELLIGENCE

**MINOR PROJECT REPORT**

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## BONAFIDE CERTIFICATE

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# ABSTRACT

Road infrastructure plays a critical role in ensuring safe and efficient transportation; however, increasing traffic density and environmental factors have led to a rise in road damage and accident risks. Traditional monitoring methods often lack efficiency, real-time insights, and integrated analysis, making it difficult for authorities to identify high-risk areas and take timely action. To address these challenges, this project presents a Road Damage and Accident Risk Analytics Dashboard developed using Business Intelligence techniques. The proposed system integrates multiple datasets, including road condition data, traffic information, weather parameters, and accident history, into a centralized analytical framework. Data preprocessing and modeling techniques are applied to ensure data consistency and reliability. The system utilizes interactive visualizations such as charts, maps, and key performance indicators to represent complex data in an intuitive and meaningful manner. A multi-page dashboard structure is designed to provide comprehensive insights through modules such as executive overview, road damage analysis, accident risk assessment, maintenance evaluation, city-level analysis, and time trend analysis. The dashboard enables stakeholders to monitor infrastructure conditions, identify high-risk zones, and evaluate maintenance performance effectively. It supports data-driven decision-making by providing actionable insights that assist in prioritizing repairs, optimizing resource allocation, and improving road safety. Overall, the project demonstrates the effectiveness of Business Intelligence tools in enhancing road infrastructure management and reducing accident risks.

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# LIST OF SYMBOLS

| ABBREVIATION | DESCRIPTION |
|---|---|
| AI | Artificial Intelligence |
| BI | Business Intelligence |
| CSV | Comma-Separated Values |
| DAX | Data Analysis Expressions |
| ETL | Extract, Transform, Load |
| ML | Machine Learning |
| KPI | Key Performance Indicator |
| UI | User Interface |

# CHAPTER 1
# INTRODUCTION

Road infrastructure is a fundamental component of modern transportation systems and plays a crucial role in economic development, connectivity, and public safety. Efficient road networks facilitate the movement of goods and people, contributing to industrial growth and societal progress. However, increasing urbanization and rising vehicle density have led to the rapid deterioration of road conditions due to heavy traffic loads, environmental factors, and inadequate maintenance practices. Issues such as potholes, surface cracks, and uneven pavements significantly affect driving conditions and increase the likelihood of accidents.

Road accidents have become a major concern, particularly in developing regions where infrastructure maintenance faces multiple challenges. A significant number of accidents are associated with poor road conditions, excessive vehicle speed, adverse weather conditions, and delayed maintenance activities. Conventional road monitoring methods rely on manual inspections and fragmented data analysis, which are time-consuming and lack real-time insights. These limitations restrict the ability of authorities to effectively identify high-risk areas and implement preventive measures.

Advancements in data analytics and visualization technologies have enabled the development of intelligent road monitoring systems. A data driven approach integrates datasets such as road damage records, traffic patterns, weather conditions, and accident history for comprehensive analysis. This project focuses on developing a Road Damage and Accident Risk Analytics Dashboard using Business Intelligence tools to transform raw data into actionable insights. The system supports informed decision making and improves maintenance planning by identifying damage patterns and highlighting critical risk factors affecting road safety.

## 1.1 OVERVIEW OF ROAD DAMAGE AND ACCIDENT RISK ANALYTICS

Road damage and accident risk analytics involves the systematic examination of infrastructure conditions and safety-related factors using structured data. This analytical approach focuses on understanding how different variables interact and contribute to road degradation and accident occurrence. By evaluating parameters such as pothole count, crack severity, traffic intensity, vehicle speed, weather conditions, and accident records, the system provides a comprehensive view of road safety.

The integration of multiple datasets plays a crucial role in this analytical process. Each dataset provides valuable information about a specific aspect of road infrastructure. For instance, weather data helps in understanding the impact of rainfall on road conditions, while traffic data provides insights into congestion and vehicle movement patterns. When combined, these datasets enable a holistic analysis of factors influencing road safety.

Business Intelligence tools facilitate the processing and visualization of large volumes of data. These tools allow users to interact with data through filters, slicers, and dynamic visual elements. As a result, complex information can be presented in an intuitive format, making it easier for users to identify trends and patterns. The ability to visualize data effectively enhances decision-making and supports efficient resource allocation.

Another important aspect of road damage analytics is the identification of relationships between variables. For example, an increase in rainfall may lead to a higher number of potholes, which in turn increases accident risk. Similarly, higher traffic density combined with excessive vehicle speed can contribute to congestion and accidents. Understanding these relationships enables authorities to implement targeted interventions.

Overall, road damage and accident risk analytics provides a structured approach to analyzing infrastructure conditions. It supports proactive maintenance planning, risk mitigation, and efficient management of transportation systems. By leveraging analytical techniques and visualization tools, the system enhances the ability to monitor road safety and improve infrastructure performance.

**Key Components**

**Data Model Integration:** The data model serves as the foundation of the analytical system by integrating multiple datasets into a unified structure. It establishes relationships between different attributes such as location, time, and road segments, enabling efficient data analysis. A well-designed data model ensures consistency, reduces redundancy, and supports complex queries. This integration allows the system to analyze multiple parameters simultaneously, providing a comprehensive view of road conditions and associated risks.

**Analytical Engine:** The analytical engine utilizes Data Analysis Expressions (DAX) to compute key performance indicators and derived metrics. These calculations include accident risk index, maintenance priority score, response time, and damage severity level. The use of DAX enables dynamic calculations that update automatically based on user interactions. This enhances the analytical capability of the system and supports real-time evaluation of performance metrics.

**Dashboard Architecture:** The dashboard is structured into multiple sections, each focusing on a specific analytical aspect. These sections include overview analysis, risk assessment, maintenance evaluation, and time-based trends. The organized structure ensures clarity and facilitates easy navigation. Each dashboard page presents data through visuals for effective interpretation.

**Hierarchical Filtering:** Hierarchical filtering allows users to explore data at different levels of granularity. This feature enables analysis from broader regions to

specific road segments, providing flexibility in data exploration. By applying filters, users can focus on particular areas of interest and examine detailed information. This enhances the depth of analysis and supports targeted decision-making.

**Visualization Techniques:** Visualization techniques play a crucial role in presenting data effectively. Charts, graphs, and maps are used to represent complex data in a simplified manner. These visual elements help in identifying patterns, trends, and anomalies quickly. Effective visualization enhances user understanding and supports data-driven decision-making.

**Key Features**

**Granular Road Condition Monitoring:** The system enables detailed assessment of road surface conditions across different regions. By analyzing parameters such as pothole count and crack severity, it helps identify areas requiring immediate attention.

**Accident Risk Identification:** Accident risk is evaluated using multiple influencing factors, including traffic density, vehicle speed, and environmental conditions. High-risk zones are highlighted for preventive action.

**Traffic and Speed Analysis:** Vehicle movement patterns are analyzed to identify congestion levels and over-speeding zones. This helps in understanding traffic behavior and its impact on road safety.

**Interactive Dashboards:** The dashboard provides an interactive environment where users can explore data through filters and visual elements. This enhances user engagement and supports detailed analysis.

**Automated Performance Indicators:** Key performance metrics are automatically calculated and displayed, enabling quick assessment of system conditions without manual computation.

**Cross-Filtering Capability:** Selections made in one visual element dynamically update all related visuals, ensuring consistent and connected analysis across the dashboard.

**Report Generation and Export:** The system allows users to generate and export reports in multiple formats, facilitating documentation and information sharing.

## 1.2 IMPORTANCE OF ROAD SAFETY ANALYTICS

Road safety analytics plays a crucial role in improving transportation systems by enabling data-driven decision-making. It provides insights into factors affecting road conditions and accident risks, allowing authorities to implement effective strategies. By identifying hazardous areas and understanding risk patterns, the system supports timely interventions and reduces accident occurrences.

Efficient maintenance planning is another significant benefit of road safety analytics. By prioritizing repairs based on severity and risk levels, resources can be utilized effectively. Real-time monitoring capabilities enable quick identification of issues and faster response to critical situations. Additionally, the system enhances accountability by providing performance metrics for maintenance activities.

The use of analytical insights also supports policy development and strategic planning. Decision-makers can evaluate the effectiveness of existing measures and implement improvements based on data-driven evidence. Overall, road safety analytics contributes to safer transportation systems and improved infrastructure management.

Furthermore, the integration of advanced analytical frameworks within road safety systems enhances the capability to perform comparative and longitudinal analysis across different regions and time periods. This enables authorities to

evaluate the effectiveness of previously implemented maintenance strategies and safety measures with greater precision. By continuously monitoring performance indicators and identifying deviations from expected patterns, the system supports adaptive decision-making and continuous improvement. In addition, the structured use of data analytics promotes transparency in infrastructure management by providing measurable outcomes and evidence-based reporting. This not only strengthens administrative efficiency but also builds public trust in transportation governance. As a result, road safety analytics serves as a critical foundation for developing sustainable, resilient, and intelligent transportation systems.

## 1.3 OBJECTIVES OF THE SYSTEM

The primary objective of this project is to develop a comprehensive analytical dashboard using Business Intelligence techniques for evaluating road damage and accident risks.

- To analyze road condition data and identify damage patterns
- To evaluate factors influencing accident probability
- To design interactive dashboards for data visualization
- To monitor maintenance performance and response efficiency
- To support informed decision-making for infrastructure management.

## 1.4 SCOPE AND LIMITATIONS

The scope of the proposed Road Damage and Accident Risk Analytics Dashboard encompasses the comprehensive analysis of structured datasets related to road infrastructure conditions, traffic characteristics, environmental parameters, and accident records. The system is designed to integrate and process data obtained from multiple sources, enabling a unified analytical framework for evaluating road safety and infrastructure performance. By leveraging Business Intelligence techniques, the

system transforms raw data into meaningful insights through interactive dashboards and visual representations.

Within the defined scope, the system facilitates the identification of damage patterns, assessment of accident risk levels, and evaluation of maintenance performance. It supports multi-dimensional analysis by incorporating variables such as pothole frequency, crack severity, traffic density, vehicle speed, weather conditions, and historical accident data. The visualization capabilities allow users to explore relationships between these variables, enabling a deeper understanding of factors influencing road safety. Additionally, the system provides functionalities such as filtering, drill-down analysis, and cross-visual interaction, which enhance the analytical experience and support informed decision-making.

The system is primarily intended for descriptive and diagnostic analysis, focusing on understanding existing conditions and identifying trends within the available data. It aids in highlighting high-risk areas, prioritizing maintenance activities, and improving resource allocation strategies. Furthermore, the dashboard serves as a decision-support tool for authorities, enabling them to monitor infrastructure performance and evaluate the effectiveness of maintenance operations.

Despite its capabilities, certain limitations must be acknowledged. The effectiveness of the system is highly dependent on the quality, completeness, and consistency of the input data. Inaccurate or incomplete datasets may lead to misleading insights and affect decision-making outcomes. The system does not incorporate real-time data acquisition mechanisms such as IoT sensors or live traffic feeds, which limits its ability to provide instantaneous updates. Additionally, predictive modeling techniques and advanced machine learning algorithms are not implemented within the current scope, restricting the system to descriptive and exploratory analysis rather than forecasting future conditions.

Another limitation is related to scalability and data integration. While the system is capable of handling structured datasets, integrating unstructured data sources or large-scale real-time streams may require additional infrastructure and computational resources. Furthermore, the system assumes a consistent data schema and may require adjustments when applied to different geographical regions or data formats.

Overall, the scope of the project is focused on providing a robust analytical framework for road damage and accident risk evaluation, while the identified limitations highlight potential areas for future enhancement and system expansion.

## 1.5 ORGANIZATION OF THE DOCUMENT

This document is organized in a structured manner to provide a clear and systematic presentation of the project. The initial section introduces the background, objectives, and significance of road damage and accident risk analytics, establishing the foundation for the study. It outlines the key challenges in road infrastructure monitoring and highlights the need for a data-driven analytical approach.

The subsequent sections focus on reviewing existing research and analytical methods, followed by a detailed discussion of current systems and their limitations. The proposed system is then presented with its architecture, features, and advantages. Further sections describe the methodology, including data collection, preprocessing, modeling, and visualization techniques used to develop the analytical dashboard.

The final sections present the implementation details and analytical results, demonstrating the effectiveness of the system. The document concludes with a summary of findings and potential future enhancements. This structured flow ensures logical progression and enhances readability, enabling a comprehensive understanding of the project.

## 1.6 TARGET AUDIENCE

The proposed Road Damage and Accident Risk Analytics Dashboard is designed to cater to a diverse group of stakeholders involved in transportation management, infrastructure planning, and data analysis. The system is intended to provide actionable insights that support decision-making across various levels of administration and technical expertise.

Government authorities and transportation departments constitute the primary audience for this system. These stakeholders are responsible for maintaining road infrastructure, ensuring public safety, and implementing policies related to transportation management. The dashboard enables them to monitor road conditions, identify high-risk areas, and prioritize maintenance activities based on data-driven insights.

Urban planners and infrastructure managers also benefit from the system by gaining a comprehensive understanding of road usage patterns and infrastructure performance. The analytical insights help in designing efficient transportation networks, optimizing resource allocation, and improving long-term planning strategies.

Data analysts and technical professionals represent another important group of users. The system provides them with a platform to explore datasets, perform analytical operations, and derive meaningful conclusions. The use of interactive visualizations enhances their ability to interpret complex data and communicate findings effectively.

Additionally, the dashboard serves as a valuable resource for researchers and academicians working in the fields of transportation engineering, data analytics, and urban development. It provides a practical example of applying Business

Intelligence techniques to real-world problems, supporting further research and innovation.

Overall, the system is designed to be user-friendly and accessible, enabling both technical and non-technical users to interact with the data and gain valuable insights.

## 1.7 ASSUMPTIONS AND DEPENDENCIES

The effective implementation of the proposed system relies on the availability of accurate and well-structured datasets, including road conditions, traffic patterns, weather information, and accident records. These datasets must be obtained from reliable sources and updated regularly to ensure meaningful analysis.

In addition, user competency in interpreting dashboards and understanding visual representations plays a significant role in deriving insights. The reliability of the system also depends on stable software infrastructure, well defined data relationships, and validated analytical measures, ensuring consistent performance and informed decision making in road infrastructure management.

Furthermore, the system's effectiveness can be enhanced through continuous data validation, periodic updates, and scalability to accommodate growing datasets. Regular monitoring and maintenance of the analytical framework ensure that the system remains accurate, responsive, and adaptable to evolving infrastructure requirements.

# CHAPTER 2
# LITERATURE SURVEY

## 2.1 VISION-BASED ROAD DAMAGE DETECTION USING DEEP LEARNING TECHNIQUES

In this paper, the authors have proposed a vision-based approach for automatic road damage detection using deep learning models. The study focuses on identifying different types of road surface damages such as potholes, cracks, and surface wear using convolutional neural networks (CNN). The model is trained on annotated image datasets and is capable of extracting spatial features effectively. The proposed method improves detection accuracy by leveraging hierarchical feature extraction and image classification techniques.

The experimental results demonstrate that deep learning models outperform traditional image processing techniques in terms of accuracy and robustness. The system also supports real-time detection when integrated with camera-based systems. However, the approach requires high-quality labeled datasets and computational resources for training. The study highlights the potential of AI-driven systems in automating road condition monitoring and reducing manual inspection efforts.

## 2.2 DEEP LEARNING FRAMEWORK FOR PAVEMENT CRACK SEGMENTATION

This research presents a deep learning-based segmentation framework for detecting pavement cracks. The authors have developed a segmentation model using advanced neural network architectures to accurately identify crack patterns in road surfaces. The model processes high-resolution images and segments damaged regions with improved precision.

The proposed approach enhances detection performance by capturing fine-grained details of cracks, which are often missed by conventional methods. The system is evaluated using benchmark datasets, showing significant improvements in segmentation accuracy. Despite its effectiveness, the method requires extensive training data and may face challenges in varying lighting and environmental conditions. The study contributes to improving maintenance planning by enabling accurate detection of surface defects.

## 2.3 TRAFFIC ACCIDENT RISK PREDICTION USING MACHINE LEARNING AND ENVIRONMENTAL FACTORS

In this paper, the authors have proposed a machine learning-based approach for predicting traffic accident risks by considering environmental and traffic-related parameters. The model integrates variables such as weather conditions, traffic density, vehicle speed, and road conditions to estimate accident probability.

The study utilizes classification algorithms to analyze historical accident data and identify patterns associated with high-risk scenarios. Experimental results indicate that environmental factors such as rainfall and visibility significantly influence accident occurrences. The model provides valuable insights for proactive safety measures and risk mitigation strategies. However, the accuracy of predictions depends on the quality and completeness of the input data.

## 2.4 INTELLIGENT POTHOLE DETECTION USING HYBRID CONVOLUTIONAL NEURAL NETWORKS

This research introduces a hybrid CNN-based approach for detecting potholes in real-time. The proposed model combines multiple convolutional layers with feature fusion techniques to improve detection accuracy. The system processes road images captured through mobile or vehicle-mounted cameras and identifies potholes with high precision.

The hybrid architecture enhances the model's ability to distinguish between different surface irregularities. The results demonstrate improved performance compared to traditional detection methods. The study also emphasizes the importance of real-time detection for timely maintenance. However, computational complexity and hardware requirements remain challenges for large-scale deployment.

## 2.5 ARTIFICIAL INTELLIGENCE-BASED ROAD CONDITION MONITORING FOR SMART CITIES

In this paper, the authors have proposed an AI-driven framework for monitoring road conditions in smart city environments. The system integrates data from multiple sources, including sensors, cameras, and traffic systems, to provide a comprehensive view of road infrastructure.

The framework uses machine learning algorithms to analyze data and detect anomalies in road conditions. The results highlight the effectiveness of AI in improving infrastructure management and reducing maintenance delays. The study also discusses scalability and integration challenges in smart city applications. The proposed system supports data-driven decision-making and enhances urban planning strategies.

## 2.6 SPATIO-TEMPORAL ANALYSIS OF ROAD ACCIDENT SEVERITY USING DEEP LEARNING

This study focuses on analyzing road accident severity using spatio-temporal deep learning models. The authors have utilized historical accident data combined with spatial and temporal features to classify accident severity levels.

The proposed model captures patterns across different locations and time intervals, enabling a deeper understanding of accident trends. The results indicate

that temporal factors such as peak traffic hours and seasonal variations significantly influence accident severity. The study provides valuable insights for traffic management and safety planning. However, the model requires large datasets and computational resources for effective training.

## 2.7 AUTOMATED ROAD SURFACE DAMAGE CLASSIFICATION USING TRANSFER LEARNING

In this paper, the authors have proposed a transfer learning-based approach for classifying road surface damages. Pre-trained deep learning models are fine-tuned using domain-specific datasets to improve classification performance.

The approach reduces training time and improves accuracy by leveraging existing knowledge from large datasets. The system classifies different types of damages such as cracks, potholes, and surface distortions. Experimental results show that transfer learning significantly enhances model performance compared to training from scratch. The study demonstrates the efficiency of transfer learning in road damage analysis.

## 2.8 REAL-TIME ROAD CRACK DETECTION USING YOLO-BASED DEEP NETWORKS

This research presents a real-time crack detection system using YOLO (You Only Look Once) deep learning architecture. The model is designed for fast and accurate object detection, making it suitable for real-time applications.

The system processes video streams and detects cracks on road surfaces with high speed and accuracy. The results indicate that YOLO-based models provide a balance between detection speed and performance. The study highlights the importance of real-time monitoring for immediate maintenance actions. However, performance may vary based on image quality and environmental conditions.

## 2.9 MACHINE LEARNING-BASED URBAN ROAD MAINTENANCE PREDICTION MODEL

In this paper, the authors have developed a predictive model for urban road maintenance using machine learning techniques. The model analyzes historical maintenance data, traffic patterns, and environmental conditions to predict future maintenance requirements.

The proposed system supports proactive maintenance planning by identifying roads that are likely to deteriorate. The results demonstrate improved efficiency in resource allocation and infrastructure management. The study emphasizes the importance of predictive analytics in reducing maintenance costs and improving road quality.

## 2.10 INTEGRATED DEEP LEARNING APPROACH FOR ROAD DAMAGE AND ACCIDENT RISK ANALYSIS

This study presents an integrated framework combining road damage detection and accident risk analysis using deep learning techniques. The system analyzes both visual and numerical data to provide a comprehensive understanding of road safety.

The model identifies damaged road segments and correlates them with accident risk factors such as traffic density and weather conditions. The results show that integrating multiple data sources improves risk prediction accuracy. The study highlights the need for combined analytical approaches to enhance road safety and infrastructure management.

# CHAPTER 3
# POWER BI

## 3.1 POWER BI INTRODUCTION

Power BI is a powerful Business Intelligence (BI) and data visualization tool developed by Microsoft. It is designed to transform raw data into meaningful insights through interactive dashboards and reports. Power BI enables users to connect to multiple data sources, process and analyze large datasets, and present the results in a visually appealing and user-friendly manner.

Power BI is widely used for data analysis, reporting, and decision-making across various industries. It provides a comprehensive platform for data integration, data modeling, and visualization, allowing organizations to gain insights and improve operational efficiency. The tool supports real-time data updates and interactive features such as filtering, slicing, and drill-down, which enhance the analytical experience.

The advantages of Power BI are as follows:

- **Data Connectivity:** Power BI supports a wide range of data sources including Excel, SQL databases, cloud services, and APIs.
- **Interactive Visualization:** It provides dynamic charts, graphs, and dashboards that allow users to explore data effectively.
- **Real-Time Analysis:** Power BI enables real-time data monitoring and updates for timely decision-making.
- **User-Friendly Interface:** It offers an intuitive drag-and-drop interface suitable for both technical and non-technical users.
- **Scalability:** Power BI can handle both small datasets and large-scale enterprise data efficiently.

## 3.2 CHARACTERISTICS OF POWER BI

- **Data Integration:** Power BI allows seamless integration of data from multiple sources into a single platform.
- **Data Transformation:** It provides tools such as Power Query to clean, transform, and preprocess raw data before analysis.
- **Data Modeling:** Users can create relationships between tables and design structured data models for efficient analysis.
- **Interactive Dashboards:** Power BI supports highly interactive dashboards with filters, slicers, and drill-through options.
- **Cloud-Based and Desktop Support:** Power BI is available as both desktop software and cloud service, enabling flexibility in usage.
- **Advanced Analytics:** It supports DAX (Data Analysis Expressions) for creating complex calculations and measures.
- **Security and Access Control:** Power BI provides role-based access and data security features to protect sensitive information.

## 3.3 POWER BI AND DATA ANALYTICS

Data analytics plays a crucial role in extracting meaningful insights from large volumes of data. Power BI serves as a powerful platform for performing data analysis by integrating data processing, modeling, and visualization in a single environment.

In modern applications, data is often collected from multiple sources and exists in structured and unstructured formats. Power BI helps in transforming this data into structured formats through preprocessing and modeling techniques. It allows users to perform descriptive and diagnostic analysis to identify patterns, trends, and anomalies.

Power BI also enables the creation of analytical dashboards that provide real-time insights into business operations. These dashboards help decision-makers monitor key performance indicators (KPIs), evaluate system performance, and identify areas for improvement. In the context of road damage and accident risk analytics, Power BI plays a vital role in analyzing infrastructure data and presenting insights for effective decision-making.

## 3.4 DATA VISUALIZATION IN POWER BI

Data visualization is the process of representing data in graphical formats such as charts, graphs, and maps. Power BI provides a wide range of visualization tools that help users understand complex data easily and effectively.

The following types of visualizations are commonly used in Power BI:

- **Bar and Column Charts:** Used to compare values across categories.
- **Line Charts:** Used to analyze trends over time.
- **Pie and Donut Charts:** Represent proportions and distributions.
- **Maps:** Visualize geographical data and location-based insights.
- **Heatmaps:** Show patterns and intensity variations.
- **KPI Cards:** Used to display key performance indicators.

Data visualization enhances the interpretation of complex datasets by presenting information in a clear and structured visual format. It transforms raw data into charts, graphs, and maps, making it easier to understand patterns, relationships, and variations within the data. By simplifying large volumes of information, it reduces analytical complexity and improves clarity. Additionally, data visualization supports effective decision making by highlighting key trends, correlations, and anomalies, enabling users to quickly derive meaningful insights and make informed decisions.

## 3.5 POWER BI IN ROAD DAMAGE AND ACCIDENT RISK ANALYSIS

Power BI serves as an effective Business Intelligence platform for analyzing and visualizing data related to road damage and accident risk. It integrates diverse datasets, including road condition records, traffic density, weather information, and historical accident data, into a unified analytical environment. By transforming heterogeneous data into structured formats, the platform enables comprehensive analysis and supports data-driven decision-making in transportation systems.

The platform offers advanced data modeling capabilities, allowing relationships to be established between parameters such as location, time, road type, and damage severity. Using Data Analysis Expressions (DAX), key performance indicators including accident risk index, maintenance priority score, and damage severity levels can be computed. In addition, interactive visualization tools such as charts, maps, heatmaps, and KPI indicators facilitate efficient interpretation of complex datasets and help identify high-risk regions.

Furthermore, Power BI supports continuous monitoring and reporting by enabling real-time updates and interactive dashboard sharing. It assists authorities in tracking infrastructure conditions, evaluating maintenance performance, and analyzing the influence of environmental and traffic-related factors. Overall, the use of Power BI enhances transparency, improves maintenance planning, and contributes to safer and more efficient road management systems.

# CHAPTER 4
# METHODOLOGY

## 4.1 EXISTING METHOD

The existing method for road damage monitoring and accident risk assessment is primarily based on conventional inspection techniques, isolated record management systems, and static reporting mechanisms. In most transportation departments, road conditions are assessed manually through field surveys, periodic inspections, and maintenance logs maintained by different authorities. These methods focus on identifying visible road defects such as potholes, cracks, uneven surfaces, and drainage issues through physical verification.

Although these traditional methods have been widely used for infrastructure monitoring, they suffer from several limitations related to scalability, integration, response efficiency, and analytical capability. Existing approaches are generally fragmented and lack centralized analytical frameworks for correlating road conditions with traffic flow, environmental factors, and accident history.

In addition to operational inefficiencies, conventional methods provide only limited support for strategic planning and long-term infrastructure management. Since the collected information is often stored in disconnected spreadsheets, manual reports, or departmental registers, extracting meaningful patterns from historical records becomes difficult. This lack of integrated analysis reduces the ability to identify recurring damage zones, seasonal deterioration trends, and accident-prone road segments. As a result, maintenance planning is frequently reactive, with corrective measures initiated only after public complaints or major failures occur.

The absence of interactive visualization and real-time performance monitoring further restricts the usefulness of existing systems. Decision-makers are unable to dynamically explore city-wise comparisons, time-based damage progression, or

maintenance backlog distribution. Such limitations reduce transparency in infrastructure governance and delay preventive interventions. To understand the drawbacks of conventional systems in a structured manner, the existing method is analyzed under the following subsections.

### 4.1.1 MANUAL ROAD CONDITION INSPECTION

The most commonly adopted existing approach involves manual inspection of roads by field engineers and maintenance staff. Inspection teams physically visit road segments and visually examine defects such as potholes, cracks, rutting, surface wear, and edge failures. The identified damages are documented in inspection sheets or departmental registers for future maintenance planning.

This process is highly dependent on human observation and judgment, which introduces inconsistency in defect classification and severity assessment. The absence of standardized digital measurements often leads to subjective evaluations, affecting the reliability of maintenance decisions.

Another major challenge is the time required to inspect large road networks. Urban and rural transportation systems consist of thousands of road segments, making manual inspection a labor-intensive and time-consuming task. As a result, many defects remain unnoticed until they become severe enough to cause accidents or traffic disruptions.

### 4.1.2 DEPARTMENT-WISE FRAGMENTED DATA STORAGE

In traditional systems, data related to road damage, accident reports, traffic density, and maintenance records are usually stored separately across multiple departments. Transportation authorities, municipal corporations, traffic police, and weather agencies often maintain their own independent databases or spreadsheets.

Because of this fragmented storage structure, integrated analysis becomes extremely difficult. For example, road damage records may not be directly linked with accident locations, making it challenging to identify whether a damaged road segment contributes to repeated accidents.

This lack of centralized data architecture prevents stakeholders from establishing meaningful relationships between variables such as:

- Road damage severity
- Traffic congestion
- Weather impact
- Vehicle speed
- Historical accident frequency
- Repair completion time

The absence of a unified analytical environment reduces the ability to derive strategic insights from available datasets.

### 4.1.3 STATIC REPORTING AND LIMITED VISUALIZATION

Another limitation of the existing method is the dependency on static reports such as printed documents, Excel sheets, and departmental summary files. These reports provide only tabular representations of data and are generally prepared periodically rather than dynamically.

Static reporting does not support interactive exploration of infrastructure data. Users cannot drill down into city-level trends, compare damage severity across regions, or identify accident hotspots visually. This creates difficulty in understanding large-scale transportation patterns and reduces the efficiency of decision-making.

Without dashboards, heatmaps, KPI indicators, or trend charts, authorities are forced to interpret raw numbers manually. This increases the complexity of identifying risk-prone zones and delays preventive action.

### 4.1.4 REACTIVE MAINTENANCE APPROACH

Traditional maintenance planning is largely reactive in nature. Repairs are usually initiated only after receiving public complaints, accident reports, or visible infrastructure failures. This delayed response mechanism affects road safety and increases maintenance costs over time.

Because the existing system does not continuously analyze damage progression, small cracks and potholes often expand into major structural failures before action is taken. Delayed intervention leads to:

- Increased accident probability
- Higher vehicle damage complaints
- Greater repair expenditure
- Reduced road lifespan
- Traffic congestion during emergency repairs

The absence of analytical prioritization models prevents authorities from categorizing roads based on urgency, severity, and accident exposure.

### 4.1.5 LACK OF MULTI-FACTOR ACCIDENT RISK ANALYSIS

Existing methods often treat accident analysis and road maintenance as separate operational domains. Accident reports are generally reviewed independently by traffic departments, while infrastructure teams focus only on physical road defects.

This separation prevents multi-factor risk analysis involving:

- Pothole count
- Crack type
- Rainfall intensity
- Visibility conditions
- Traffic level
- Speed violations
- Surface age
- Previous accident history

Without combining these attributes, authorities cannot accurately assess accident risk levels for specific road segments. As a result, preventive safety planning remains limited.

The lack of correlation-based analysis also makes it difficult to identify recurring causes behind accidents in particular zones.

### 4.1.6 LIMITATIONS OF THE EXISTING METHOD

The conventional existing method presents several operational and analytical limitations that significantly affect the efficiency of transportation safety management and infrastructure monitoring. The dependence on manual procedures, disconnected data repositories, and non-interactive reporting structures reduces the ability of stakeholders to respond effectively to road safety challenges. These constraints also limit the scope of comparative analysis, timely maintenance decisions, and long-term planning.

**Major Limitations**

- Manual inspection consumes significant time and manpower
- High probability of human error in damage classification

- No centralized integration of road, traffic, and accident datasets
- Static reports restrict deep analytical exploration
- Lack of predictive or trend-based maintenance planning
- Delayed response to critical infrastructure failures
- No real-time KPI monitoring
- Difficult to identify city-wise accident hotspots
- Poor support for strategic resource allocation.

These limitations collectively highlight the inefficiency of traditional road monitoring frameworks in handling the growing complexity of modern transportation systems. As road networks expand and traffic density increases, reliance on manual inspection and isolated reporting mechanisms becomes increasingly unsustainable. The absence of centralized analytical integration prevents meaningful correlation between road condition indicators, traffic behavior, environmental parameters, and accident history, thereby reducing the effectiveness of risk identification and maintenance prioritization. Furthermore, static reporting structures fail to provide stakeholders with dynamic insights required for timely intervention, trend monitoring, and performance evaluation. Such constraints not only delay corrective actions but also impact policy formulation, budget optimization, and infrastructure sustainability. Hence, the need for an intelligent and interactive Business Intelligence–based analytical framework becomes essential for ensuring proactive road safety management, efficient maintenance governance, and long-term transportation planning for sustainable strategic decision support. Furthermore, such a framework facilitates data-driven insights and real-time monitoring, enabling authorities to respond effectively to road conditions and emerging risks.

## 4.2 PROPOSED METHOD

The proposed method introduces a centralized Business Intelligence–based analytical framework for monitoring road damage severity, evaluating accident risk levels, and assessing maintenance performance using Power BI. The system is designed to overcome the limitations of conventional inspection-driven methods by integrating multiple structured datasets into a unified decision-support environment. The collected data includes road segment details, pothole count, crack type, traffic density, weather conditions, speed indicators, accident history, maintenance priority, repair status, and reporting date. These heterogeneous datasets are consolidated into a centralized format to support multidimensional analysis across geographic, temporal, and operational perspectives. Through systematic preprocessing, duplicate records, missing values, and inconsistent labels are corrected, while numerical and categorical fields are standardized to improve analytical reliability and dashboard responsiveness.

After preprocessing, the refined dataset is transformed into a relational Power BI data model where logical relationships are established among road condition indicators, accident records, maintenance logs, and time-based fields. Using DAX measures and calculated columns, key performance indicators such as total damage, accident risk score, average response time, pending repairs, and city-level severity trends are derived for analytical evaluation. The final output is implemented as a six-page interactive dashboard consisting of Executive Overview, Road Damage Analysis, Accident Risk Analysis, Maintenance Analysis, City-Level Insights, and Time Trend Analysis. This proposed methodology enables transportation authorities to identify critical road segments, analyze accident hotspots, optimize maintenance planning, and allocate resources strategically through data-driven insights, thereby improving road safety management and infrastructure governance.

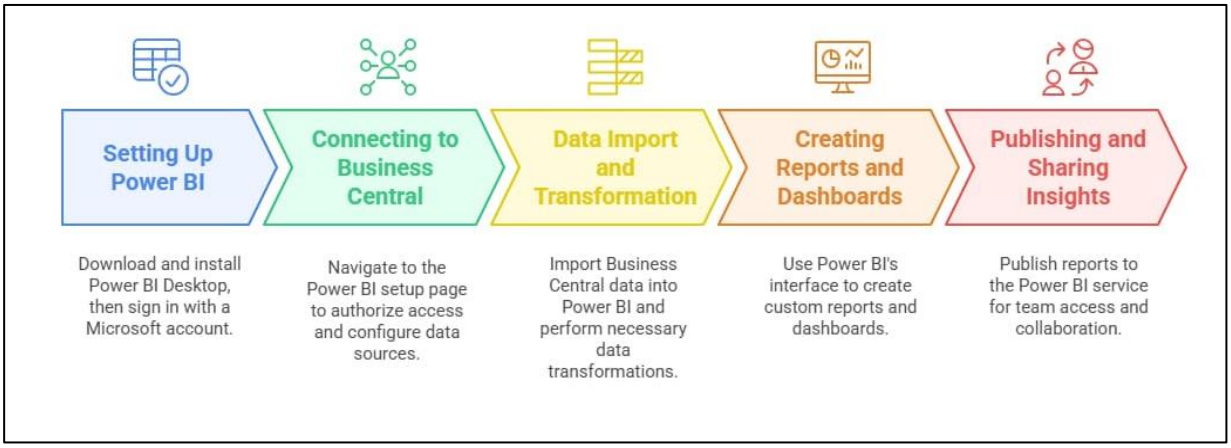



***Figure 4.2 Power BI Workflow***

### 4.2.1 DATA COLLECTION AND INTEGRATION

The first stage of the proposed method focuses on collecting structured transportation datasets from reliable and domain-relevant sources. The collected attributes include road condition indicators, traffic information, environmental factors, accident history, maintenance details, and geographic parameters. These datasets are integrated into a centralized tabular structure, enabling correlation analysis between road damage severity and accident risk conditions.

The integration process ensures that all operational and safety-related variables are available within a single analytical ecosystem, thereby improving reporting consistency and reducing fragmentation.

### 4.2.2 DATA PREPROCESSING AND QUALITY IMPROVEMENT

The second stage emphasizes improving data quality and analytical consistency through preprocessing operations. Duplicate entries, null values, invalid labels, and inconsistent field formats are corrected during this stage. Numerical attributes such as pothole count, crack count, rainfall, and response time are standardized to maintain uniformity across records.

Categorical fields including traffic level, weather condition, road type, and repair status are normalized to improve filtering and comparative analysis.

### 4.2.3 DATA MODELING IN POWER BI

After preprocessing, the refined dataset is transferred into the Power BI data model, where relationships are established among road segments, accident history, maintenance logs, and time-based fields. The modeling structure follows a relational approach that supports hierarchical analysis from **state → city → road segment level**.

Calculated measures and DAX expressions are created to derive performance metrics, risk indicators, and maintenance intelligence, enabling efficient dashboard interactivity.

### 4.2.4 MULTI-PAGE DASHBOARD IMPLEMENTATION

The proposed methodology is implemented as a six-page dashboard framework, where each page focuses on a dedicated analytical perspective. The dashboard pages include Executive Overview, Road Damage Analysis, Accident Risk Analysis, Maintenance Analysis, City-Level Insights, and Time Trend Analysis. This modular design improves readability, supports role-based access, and enables stakeholders to focus on specific transportation performance dimensions. By separating strategic, operational, and regional insights into dedicated views, the dashboard minimizes information overload and improves analytical clarity.

In addition, each dashboard page is designed with interactive slicers, filters, drill-down navigation, and KPI indicators to support detailed data exploration. Users can analyze location-specific damage trends, identify accident-prone zones, monitor maintenance delays, and evaluate seasonal infrastructure variations with greater precision. This structured visualization framework enhances user experience,

accelerates decision-making, and strengthens transportation safety monitoring across multiple administrative levels.

### 4.2.5 DECISION-SUPPORT AND STRATEGIC OUTCOME

The final stage converts dashboard insights into actionable decision-support intelligence. The system enables transportation departments to identify severe road damage locations, analyze accident hotspots, track maintenance efficiency, and allocate repair resources based on severity and urgency.

The proposed method strengthens operational transparency, improves infrastructure governance, and supports long-term transportation safety planning through evidence-based insights.

### 4.2.6 PERFORMANCE EVALUATION

The performance evaluation phase focuses on assessing the effectiveness of the proposed dashboard in supporting road damage monitoring, accident risk assessment, and maintenance management. Key indicators such as response time, repair completion status, damage severity trends, and city-wise maintenance efficiency are analyzed to measure the operational success of the system. The interactive analytical environment enables stakeholders to compare performance across multiple regions, identify delays in repair activities, and evaluate whether maintenance priorities are aligned with actual infrastructure risk levels.

The evaluation phase assesses the reliability of dashboard insights in supporting strategic decision-making. Consistency of KPI outputs, accurate cross-page filtering, and effective time-based trend visualization validate the analytical robustness of the framework. This analysis ensures continuous system improvement and supports long-term transportation planning and safety enhancement.

## 4.3 SYSTEM ARCHITECTURE

The system architecture of the Road Damage and Accident Risk Analytics Dashboard using Power BI is designed as a structured Business Intelligence framework that supports the complete flow of transportation data from collection to visualization. The architecture integrates road damage indicators, accident history, traffic parameters, environmental conditions, and maintenance records into a centralized analytical platform, enabling efficient monitoring, reporting, and decision-support for transportation authorities.

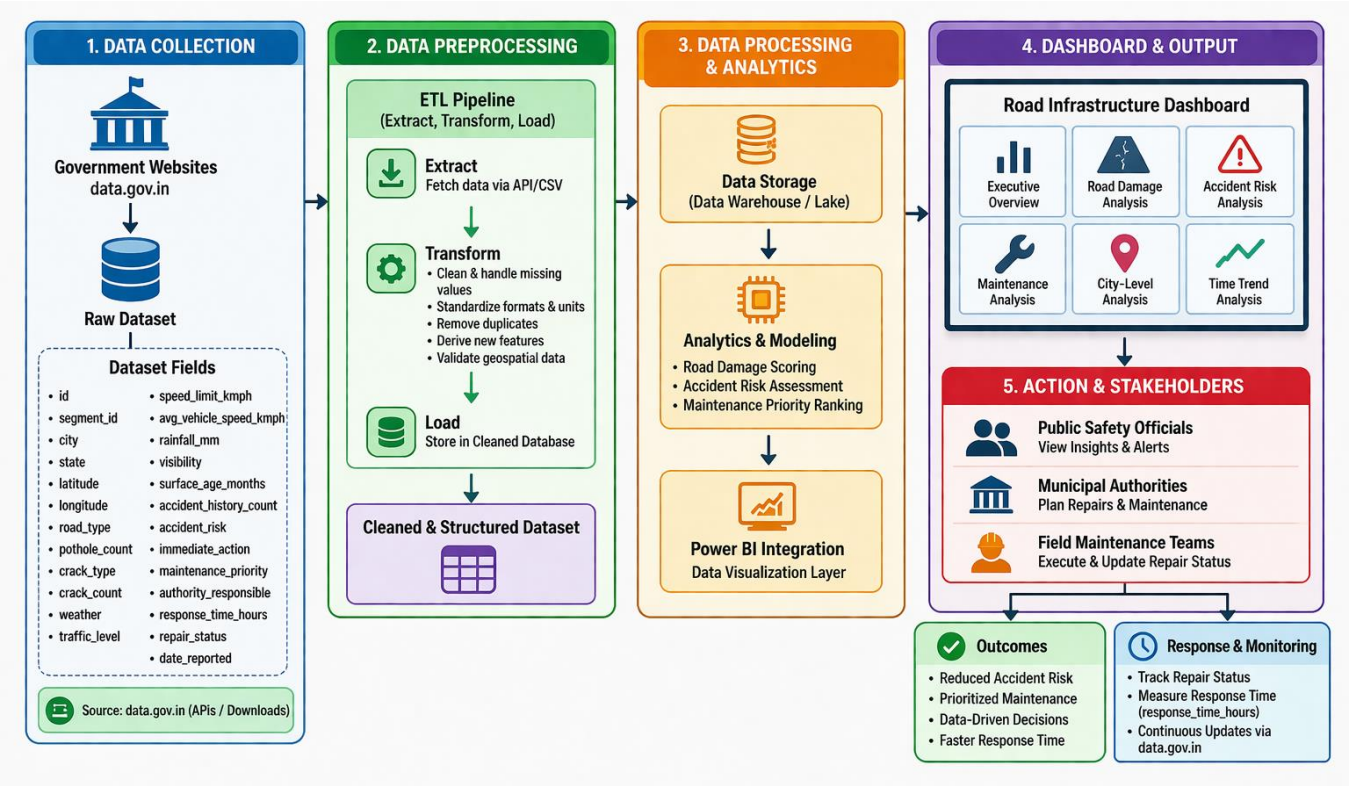



***Fig 4.3 Overall Architecture***

The architecture begins with the data collection and preprocessing stage, where raw transportation datasets are gathered from structured sources containing road segment details, pothole count, crack severity, traffic density, weather conditions, accident history, maintenance priority, and repair status. During preprocessing, duplicate records, missing values, and inconsistent labels are corrected to improve

data quality. This stage ensures that the dataset is transformed into a reliable and analysis-ready format suitable for Power BI integration.

The processed data is then transferred into the Power BI data modeling environment, where relationships are established among geographic attributes, road condition indicators, maintenance records, and time-based reporting fields. Using relational modeling and DAX measures, important metrics such as total damage, accident risk level, average response time, pending repair count, and city-wise severity trends are calculated. This structured modeling process improves analytical performance and supports seamless cross-filtering across multiple dashboard pages.

The final stage of the architecture focuses on interactive visualization and decision-support generation. The modeled data is presented through six dashboard pages including Executive Overview, Road Damage Analysis, Accident Risk Analysis, Maintenance Analysis, City-Level Insights, and Time Trend Analysis. Through charts, maps, KPI cards, slicers, and drill-down filters, the system converts complex transportation data into actionable insights. This architecture enables authorities to identify high-risk road segments, monitor maintenance efficiency, and make data-driven strategic decisions for safer and more sustainable transportation infrastructure.

In addition to its analytical capabilities, the proposed architecture provides scalability, flexibility, and efficient stakeholder accessibility. The layered flow of data from acquisition to visualization ensures smooth system performance and supports future enhancements such as real-time IoT data integration, predictive risk modeling, and smart city transportation monitoring. This makes the architecture highly suitable for modern infrastructure management systems that require continuous monitoring, transparent reporting, and evidence-based policy planning.

# CHAPTER 5
# REQUIREMENTS

## 5.1 SOFTWARE REQUIREMENTS

- Windows 11 - OS
- Power BI 2.146.705.0 (64-bit)

## 5.2 HARDWARE REQUIREMENTS

- Intel i5 Processor
- 4 GB RAM
- 2 GB Free Disk Space

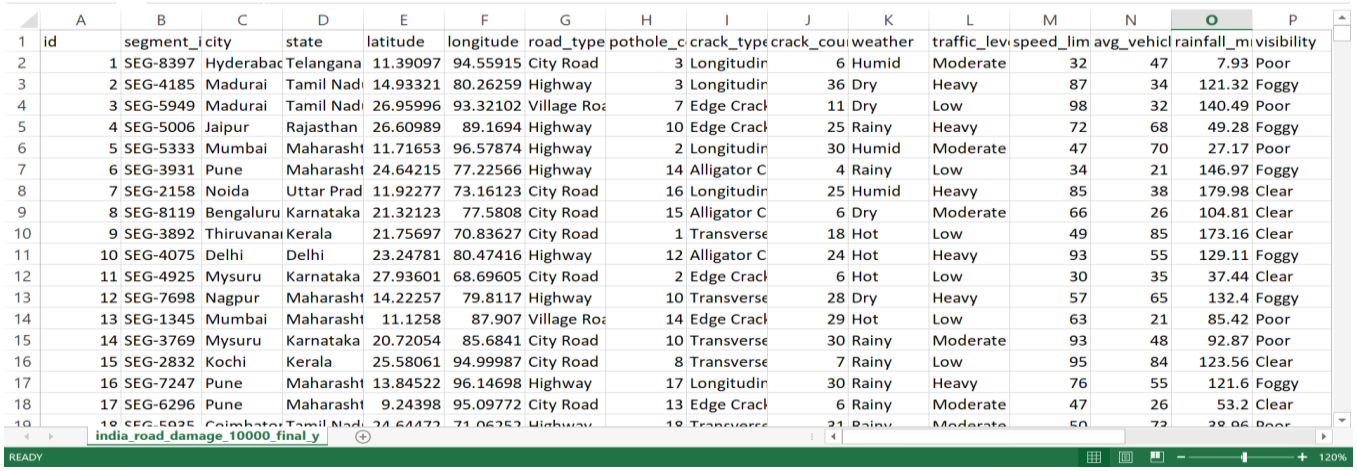
# CHAPTER 6
# RESULTS AND DISCUSSIONS

The implementation of the Road Damage and Accident Risk Analytics Dashboard demonstrates the practical effectiveness of Business Intelligence techniques in transforming complex transportation datasets into meaningful and actionable insights. The system successfully integrates data related to road infrastructure conditions, traffic characteristics, weather parameters, accident history, maintenance records, and city-level performance indicators into a unified analytical environment. Through systematic preprocessing, relational modeling, and interactive visualization, the dashboard converts raw transportation data into structured intelligence that supports comprehensive monitoring of road safety and infrastructure quality.

The results obtained from the six-page dashboard framework reveal significant analytical patterns in road damage severity, accident risk distribution, maintenance response efficiency, and regional infrastructure performance. The Executive Overview page provides high-level KPI insights such as total damage count, accident risk score, pending repair status, and average response time, enabling quick strategic assessment. The Road Damage Analysis and Accident Risk Analysis pages highlight severe pothole zones, crack distribution trends, speed-related accident risks, and environmental influences such as rainfall and visibility. Similarly, the Maintenance Analysis and City-Level Insights pages support comparative evaluation of repair performance, authority efficiency, and location-based damage intensity, while the Time Trend Analysis page effectively captures monthly and seasonal variations in transportation safety conditions.

The discussion of results confirms that the proposed Power BI dashboard significantly improves decision-making capability by enabling stakeholders to identify accident-prone segments, prioritize maintenance activities, and monitor

operational performance through dynamic visual exploration. Interactive slicers, drill-down filters, KPI cards, maps, and trend visualizations enhance the usability of the system and allow deeper analytical interpretation across multiple administrative levels. Overall, the developed system effectively supports proactive road safety management, efficient infrastructure maintenance, and evidence-based transportation planning.



| id | segment_i | city | state | latitude | longitude | road_type | pothole_c | crack_type | crack_cou | weather | traffic_lev | speed_lim | avg_vehicl | rainfall_mi | visibility |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | SEG-8397 | Hyderabac | Telangana | 11.39097 | 94.55915 | City Road | 3 | Longitudin | 6 | Humid | Moderate | 32 | 47 | 7.93 | Poor |
| 2 | SEG-4185 | Madurai | Tamil Nad | 14.93321 | 80.26259 | Highway | 3 | Longitudin | 36 | Dry | Heavy | 87 | 34 | 121.32 | Foggy |
| 3 | SEG-5949 | Madurai | Tamil Nad | 26.95996 | 93.32102 | Village Roa | 7 | Edge Crack | 11 | Dry | Low | 98 | 32 | 140.49 | Poor |
| 4 | SEG-5006 | Jaipur | Rajasthan | 26.60989 | 89.1694 | Highway | 10 | Edge Crack | 25 | Rainy | Heavy | 72 | 68 | 49.28 | Foggy |
| 5 | SEG-5333 | Mumbai | Maharasht | 11.71653 | 96.57874 | Highway | 2 | Longitudin | 30 | Humid | Moderate | 47 | 70 | 27.17 | Poor |
| 6 | SEG-3931 | Pune | Maharasht | 24.64215 | 77.22566 | Highway | 14 | Alligator C | 4 | Rainy | Low | 34 | 21 | 146.97 | Foggy |
| 7 | SEG-2158 | Noida | Uttar Prad | 11.92277 | 73.16123 | City Road | 16 | Longitudin | 25 | Humid | Heavy | 85 | 38 | 179.98 | Clear |
| 8 | SEG-8119 | Bengaluru | Karnataka | 21.32123 | 77.5808 | City Road | 15 | Alligator C | 6 | Dry | Moderate | 66 | 26 | 104.81 | Clear |
| 9 | SEG-3892 | Thiruvanai | Kerala | 21.75697 | 70.83627 | City Road | 1 | Transverse | 18 | Hot | Low | 49 | 85 | 173.16 | Clear |
| 10 | SEG-4075 | Delhi | Delhi | 23.24781 | 80.47416 | Highway | 12 | Alligator C | 24 | Hot | Heavy | 93 | 55 | 129.11 | Foggy |
| 11 | SEG-4925 | Mysuru | Karnataka | 27.93601 | 68.69605 | City Road | 2 | Edge Crack | 6 | Hot | Low | 30 | 35 | 37.44 | Clear |
| 12 | SEG-7698 | Nagpur | Maharasht | 14.22257 | 79.8117 | Highway | 10 | Transverse | 28 | Dry | Heavy | 57 | 65 | 132.4 | Foggy |
| 13 | SEG-1345 | Mumbai | Maharasht | 11.1258 | 87.907 | Village Roa | 14 | Edge Crack | 29 | Hot | Low | 63 | 21 | 85.42 | Poor |
| 14 | SEG-3769 | Mysuru | Karnataka | 20.72054 | 85.6841 | City Road | 10 | Transverse | 30 | Rainy | Moderate | 93 | 48 | 92.87 | Poor |
| 15 | SEG-2832 | Kochi | Kerala | 25.58061 | 94.99987 | City Road | 8 | Transverse | 7 | Rainy | Low | 95 | 84 | 123.56 | Clear |
| 16 | SEG-7247 | Pune | Maharasht | 13.84522 | 96.14698 | Highway | 17 | Longitudin | 30 | Rainy | Heavy | 76 | 55 | 121.6 | Foggy |
| 17 | SEG-6296 | Pune | Maharasht | 9.24398 | 95.09772 | City Road | 13 | Edge Crack | 6 | Rainy | Moderate | 47 | 26 | 53.2 | Clear |

*Figure 6.1 - Features in a .csv file*

**Figure 6.1** shows the structured input dataset used for developing the analytical dashboard. It includes key attributes required for preprocessing, data modeling, and visualization. It ensures data consistency and supports accurate analysis.

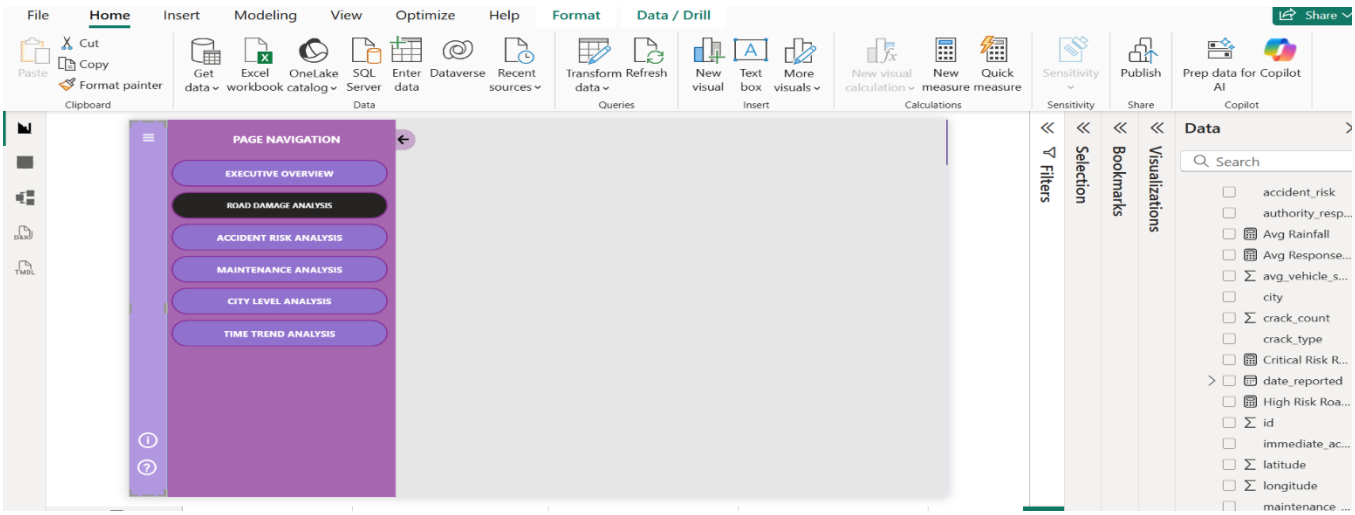



*Figure 6.2 – Page Navigation*

**Figure 6.2** presents the page navigation interface designed to provide efficient access to all six dashboard pages. It enables smooth movement between different modules and report views, allowing users to explore insights without interruption. The clear navigation structure enhances dashboard usability and supports efficient data exploration.

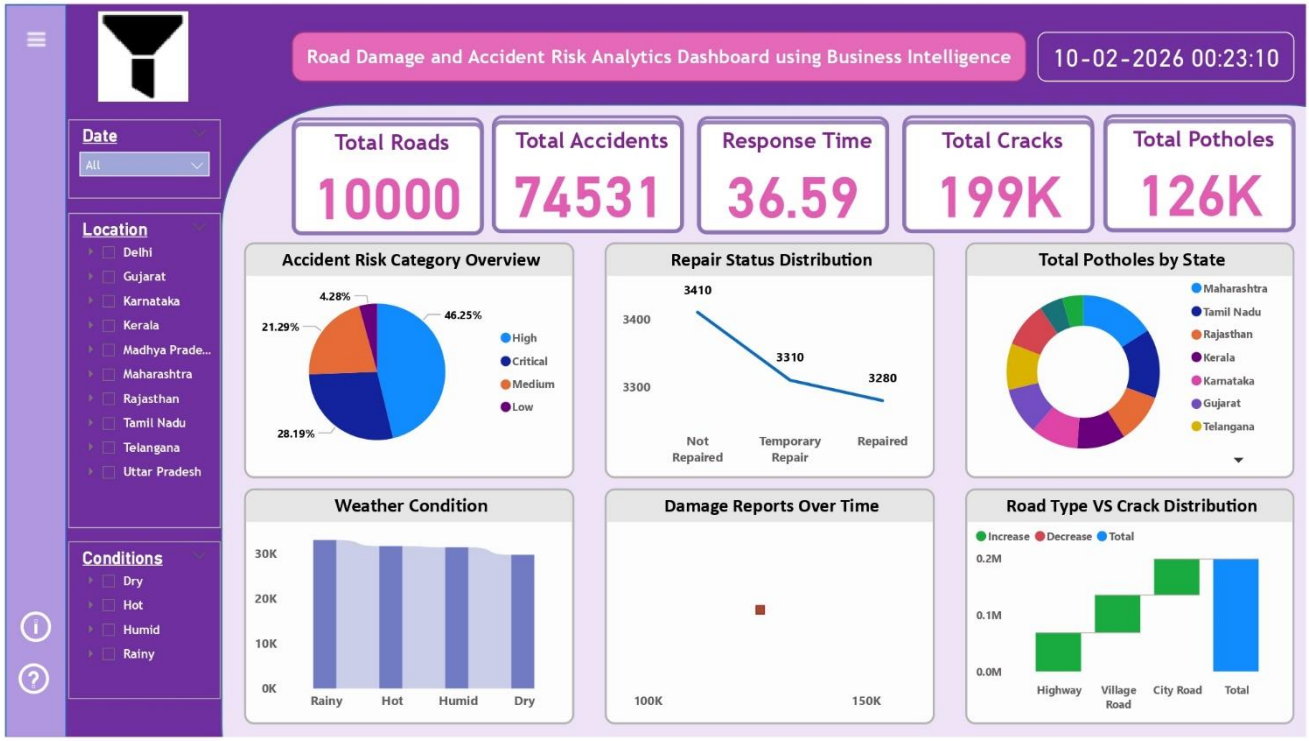



*Figure 6.3 – Executive Overview*

**Figure 6.3** shows the dashboard's summary page with key performance indicators, including total damage, accident risk, maintenance status, and response efficiency. It provides a high-level view for quick understanding and supports faster, data-driven decision-making. The page combines charts, gauges, and tables to monitor metrics across regions and road types, enabling proactive maintenance planning and prioritization of repair activities, thereby improving urban road safety and operational efficiency.

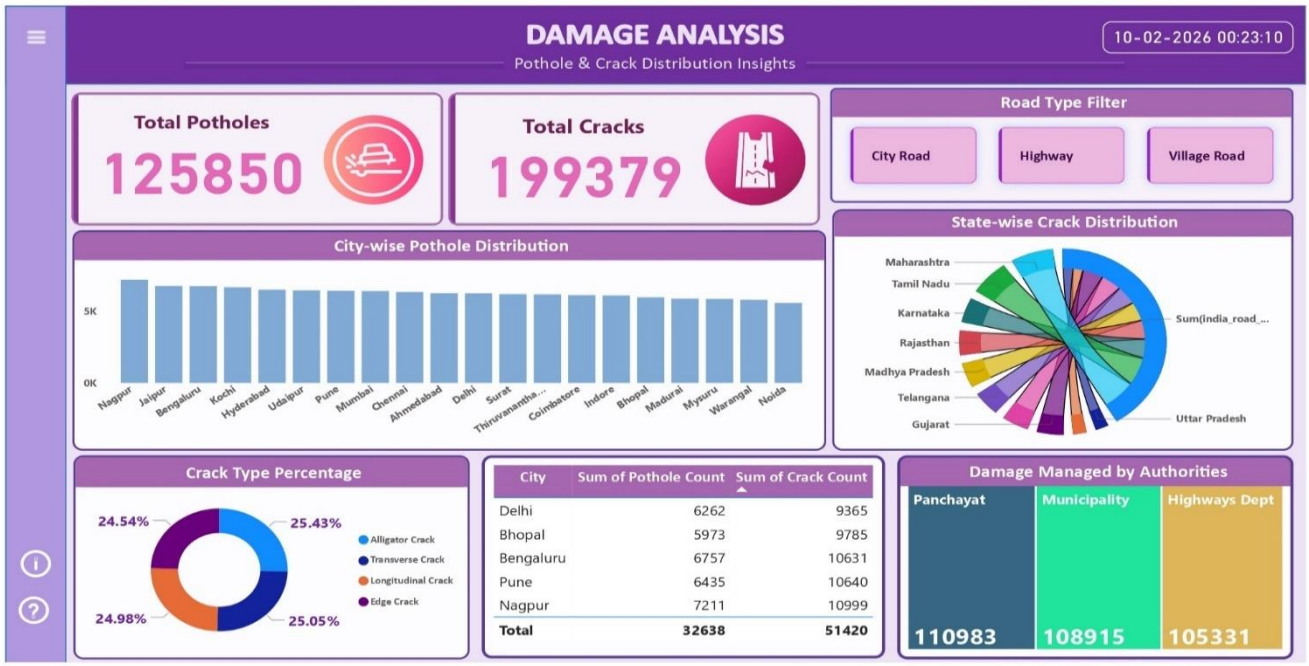



*Figure 6.4 – Road Damage Analysis*

**Figure 6.4** visualizes the distribution of potholes and cracks across different regions. It highlights areas with high damage concentration and provides insights into damage severity patterns, supporting effective maintenance planning and prioritization.

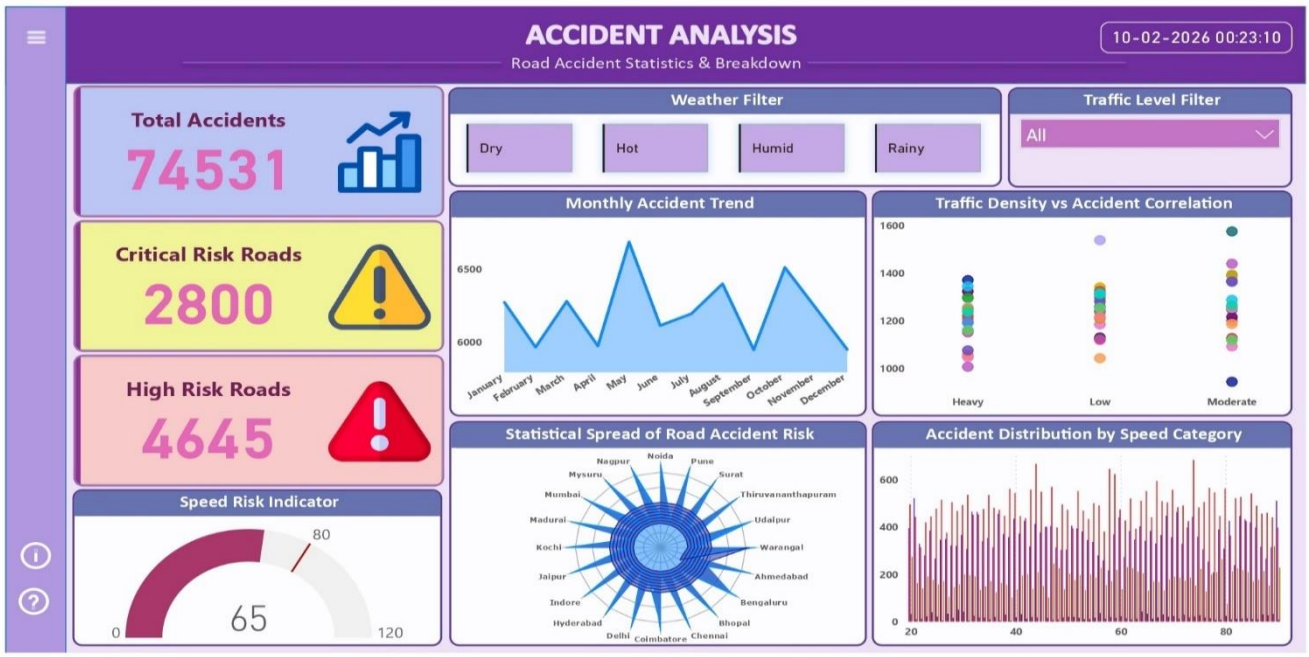
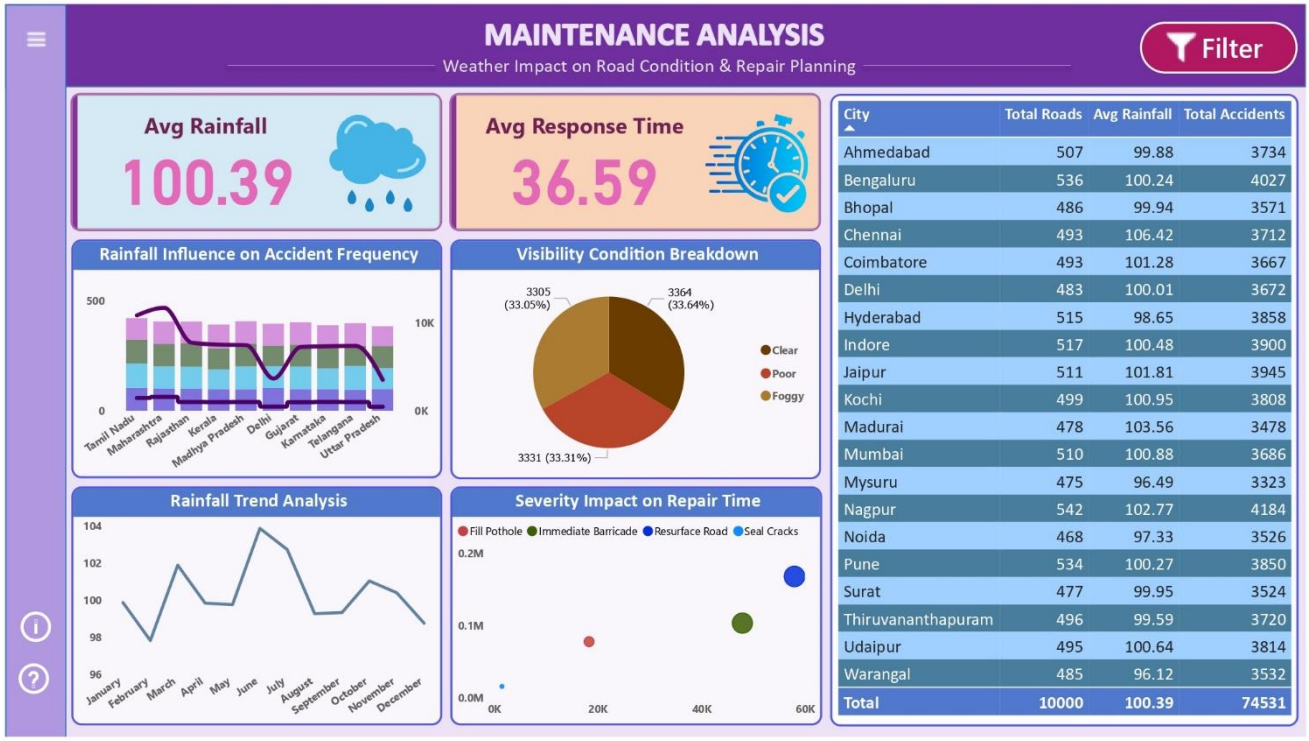



*Figure 6.5 – Accident Risk Analysis*

**Figure 6.5** presents the relationship between road damage, traffic conditions, and accident probability. It highlights high-risk areas based on speed, traffic density, weather, and historical accident data. The analysis supports safety improvement and risk reduction strategies, enabling authorities to prioritize interventions effectively.




| City | Total Roads | Avg Rainfall | Total Accidents |
|---|---|---|---|
| Ahmedabad | 507 | 99.88 | 3734 |
| Bengaluru | 536 | 100.24 | 4027 |
| Bhopal | 486 | 99.94 | 3571 |
| Chennai | 493 | 106.42 | 3712 |
| Coimbatore | 493 | 101.28 | 3667 |
| Delhi | 483 | 100.01 | 3672 |
| Hyderabad | 515 | 98.65 | 3858 |
| Indore | 517 | 100.48 | 3900 |
| Jaipur | 511 | 101.81 | 3945 |
| Kochi | 499 | 100.95 | 3808 |
| Madurai | 478 | 103.56 | 3478 |
| Mumbai | 510 | 100.88 | 3686 |
| Mysuru | 475 | 96.49 | 3323 |
| Nagpur | 542 | 102.77 | 4184 |
| Noida | 468 | 97.33 | 3526 |
| Pune | 534 | 100.27 | 3850 |
| Surat | 477 | 99.95 | 3524 |
| Thiruvananthapuram | 496 | 99.59 | 3720 |
| Udaipur | 495 | 100.64 | 3814 |
| Warangal | 485 | 96.12 | 3532 |
| **Total** | **10000** | **100.39** | **74531** |

***Figure 6.6 – Maintenance Analysis***

**Figure 6.6** illustrates a maintenance dashboard analyzing the effect of weather on road safety and repair planning. Key metrics such as average rainfall and response time provide operational insights, while visualizations of accident frequency, visibility conditions, and rainfall trends help identify high-risk areas and periods, enabling efficient resource allocation.

The bubble chart shows repair severity against time and cost for tasks like pothole filling and road resurfacing. A city-wise table summarizes total roads, rainfall, and accidents for comparative analysis. This integrated framework

improves infrastructure monitoring, maintenance planning, and proactive hazard mitigation.

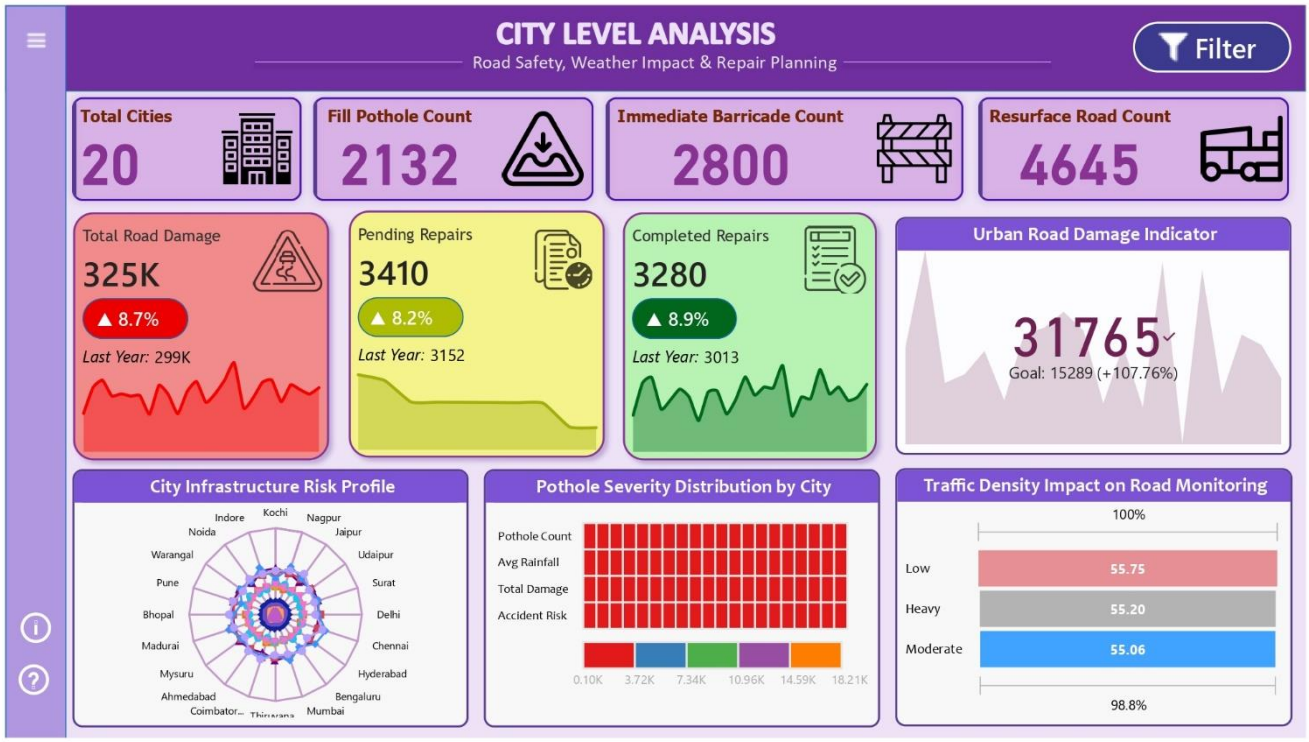



***Figure 6.7 – City Level Analysis***

**Figure 6.7** provides a consolidated overview of road damage, repair status, and traffic impact across multiple cities. It highlights key metrics such as total cities, pothole count, immediate barricades, resurfaced roads, as well as indicators like total road damage, pending repairs, and completed repairs, enabling quick assessment of infrastructure conditions and maintenance performance.

The dashboard also includes comparative visualizations such as a city risk profile radar chart, pothole severity heatmap, and traffic density impact chart. These features support city-level comparison, identification of high-risk areas, and informed decision-making for effective repair prioritization and resource allocation.

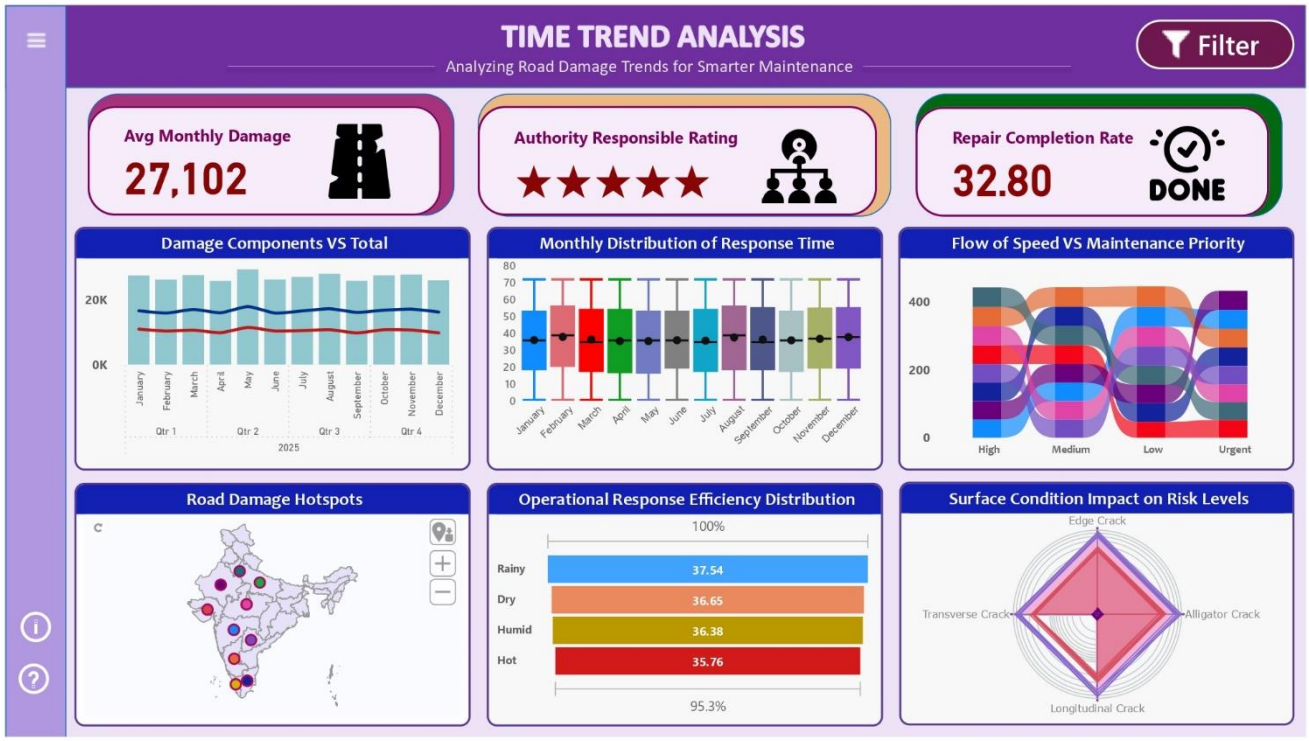



***Figure 6.8 – Time Trend Analysis***

**Figure 6.8** illustrates the Time Trend Analysis dashboard, providing insights into road damage patterns and maintenance efficiency. Key metrics such as average monthly damage, authority rating, and repair completion rate are displayed, offering a quick operational overview.

The dashboard integrates multiple visualizations, including damage component trends, response time distribution, speed versus maintenance priority flow, hotspot mapping, operational efficiency, and surface condition impact on risk levels. This comprehensive view enables data-driven monitoring, trend identification, and prioritization of maintenance activities for improved infrastructure management.

# CHAPTER 7

# CONCLUSION AND FUTURE ENHANCEMENTS

## 7.1 CONCLUSION

The Road Damage and Accident Risk Analytics Dashboard developed using Business Intelligence techniques provides an effective solution for analyzing road conditions and evaluating accident risks in a structured and data-driven manner. The system integrates multiple datasets related to road infrastructure, traffic patterns, weather conditions, and accident history into a centralized analytical platform. By transforming raw data into meaningful insights through interactive visualizations, the dashboard enhances the ability of stakeholders to understand complex relationships and identify critical factors affecting road safety.

The implementation of the system highlights the importance of data analytics in improving transportation management and infrastructure maintenance. Through visualization techniques such as charts, maps, and KPI indicators, users can monitor road damage patterns, assess accident risk levels, and evaluate maintenance performance efficiently. The multi-page dashboard structure ensures comprehensive analysis by providing dedicated views for executive overview, road damage analysis, accident risk assessment, maintenance evaluation, city-level comparison, and time-based trend analysis.

The system supports informed decision-making by enabling authorities to identify high-risk areas and prioritize maintenance activities. Despite limitations related to data dependency and the absence of real-time integration, the system establishes a strong foundation for analytical evaluation of road safety and infrastructure performance, contributing to improved efficiency and strategic planning.

## 7.2 FUTURE ENHANCEMENTS

The proposed system provides a strong analytical foundation for road damage and accident risk evaluation; however, several enhancements can be incorporated to further improve its functionality and effectiveness. One significant enhancement is the integration of real-time data sources such as IoT sensors, GPS devices, and live traffic feeds. This would enable continuous monitoring of road conditions and traffic patterns, allowing faster identification of critical issues and improving response efficiency.

Another important improvement involves the incorporation of predictive analytics and machine learning techniques. By utilizing historical data, the system can be extended to forecast potential road damage and accident risks. Predictive models such as time-series analysis and regression techniques can assist in proactive maintenance planning and accident prevention, thereby enhancing overall road safety.

The system can also be enhanced through the integration of image processing and computer vision techniques for automated road damage detection. Advanced models such as convolutional neural networks and object detection algorithms can analyze images captured from cameras or drones to identify potholes and cracks with higher accuracy. This approach reduces manual effort and improves detection efficiency.

Additionally, the dashboard can be enhanced by incorporating mobile accessibility and improved user interface design. These enhancements aim to evolve the system into a more intelligent, scalable, and efficient solution for road infrastructure management and accident risk reduction.

# REFERENCES


[1] Ahmed, M., Rahman, T., & Islam, N., "Machine Learning-Based Urban Road Maintenance Prediction Model," IEEE Sensors Journal, vol. 25, no. 1, pp. 334–345, 2025.

[2] Brown, T., Wilson, J., & Taylor, K., "Deep Neural Network Approach for Integrated Road Damage and Accident Risk Analysis," IEEE Transactions on Intelligent Transportation Systems, vol. 26, no. 4, pp. 5678–5689, 2025.

[3] Chen, X., Liu, Y., & Wang, H., "Traffic Accident Risk Prediction Using Machine Learning and Environmental Factors," IEEE Transactions on Intelligent Transportation Systems, vol. 24, no. 3, pp. 2345–2356, 2023.

[4] Gupta, D., Mehta, R., & Sharma, A., "Artificial Intelligence-Based Road Condition Monitoring for Smart Cities," IEEE Access, vol. 11, pp. 9876–9885, 2023.

[5] Huang, L., Sun, Z., & Liu, J., "Road Anomaly Detection Using LiDAR and Deep Learning," IEEE Transactions on Intelligent Transportation Systems, vol. 27, no. 2, pp. 1234–1245, 2025.

[6] Kumar, R., Singh, P., & Verma, S., "Intelligent Pothole Detection Using Hybrid Convolutional Neural Networks," IEEE Sensors Journal, vol. 23, no. 8, pp. 6789– 6798, 2023.

[7] Lee, J., Park, H., & Kim, S., "Automated Road Surface Damage Classification Using Transfer Learning," IEEE Transactions on Intelligent Vehicles, vol. 9, no. 2, pp. 145–156, 2024.

[8] Li, Y., Chen, X., & Zhao, W., “Vision-Based Road Damage Detection Using Deep Learning Techniques,” IEEE Access, vol. 10, pp. 12345–12356, 2022.

[9] Liu, R., Zhao, Q., & Wang, Y., “Spatio-Temporal Traffic Flow Prediction Using Deep Learning,” IEEE Transactions on Intelligent Transportation Systems, vol. 28, no. 1, pp. 234–245, 2026.

[10] Mohamed, A., Singh, R., & Patel, D., “Weather-Aware Traffic Risk Assessment Using Machine Learning,” IEEE Transactions on Intelligent Transportation Systems, vol. 27, no. 3, pp. 345–356, 2025.

[11] Patel, J., Nair, S., & Rao, P., “Spatio-Temporal Analysis of Road Accident Severity Using Deep Learning,” in Proc. IEEE Int. Conf. Data Science and Engineering, pp. 112–118, 2024.

[12] Sharma, A., Gupta, D., & Mehta, R., “Deep Learning for Real-Time Traffic Congestion Prediction,” IEEE Access, vol. 13, pp. 11321–11332, 2025.

[13] Wang, L., Zhao, Q., & Li, F., “Real-Time Road Crack Detection Using YOLO- Based Deep Networks,” IEEE Access, vol. 12, pp. 22345–22355, 2024.

[14] Zhang, K., Liu, H., & Wang, Y., “Deep Learning Framework for Pavement Crack Segmentation,” IEEE Transactions on Intelligent Transportation Systems, vol. 23, no. 5, pp. 4567–4578, 2022.

[15] Zhou, H., Xu, P., & Tan, J., “Multimodal Sensor Fusion for Road Damage and Traffic Incident Detection,” IEEE Sensors Journal, vol. 24, no. 4, pp. 789–798, 2024.

# APPENDIX I

## DAX FORMULAS & CALCULATIONS

### 1. TOTAL DAMAGE

Aggregates all cracks and potholes reported in the system. Provides a single metric to understand road degradation severity. Supports prioritization of maintenance and resource allocation.

```
Total_Damage =
SUM('Road Damage Detection'[crack_count]) +
SUM('Road Damage Detection'[pothole_count])
```

### 2. TOTAL RECORDS

Counts all rows in the dataset, representing the total number of reported road damage incidents. Useful for evaluating dataset scale and completeness.

```
Total Records =
COUNTROWS('Road Damage Detection')
```

### 3. TOTAL CITIES

Calculates the number of unique cities with reported road damage. Helps visualize the geographic spread of incidents and supports region-specific planning.

```
Total Cities = DISTINCTCOUNT('Road Damage Detection'[City])
```

**4. AVERAGE MONTHLY DAMAGE**

Divides total damage by the number of unique months with reported incidents. Supports trend analysis, seasonal planning, and resource allocation.

```
Avg Monthly Damage =
DIVIDE(
   [Total Damage],
   DISTINCTCOUNT('Road Damage Detection'[date_reported].[Month])
)
```

**5. COMPLETED REPAIRS**

Counts the number of incidents marked as repaired. Provides insights into maintenance performance.

```
Completed Repairs =
CALCULATE(
   COUNT('Road Damage Detection'[repair_status]),
   'Road Damage Detection'[repair_status] = "Repaired"
)
```

**6. PENDING REPAIRS**

Counts all incidents that are not fully repaired, including temporary fixes. Enables authorities to monitor backlog and prioritize urgent maintenance.

```
Pending Repairs =
CALCULATE(
   COUNT('Road Damage Detection'[repair_status]),
   'Road Damage Detection'[repair_status] = "Not Repaired")
```

```
+
CALCULATE(
    COUNT('Road Damage Detection'[repair_status]),
    'Road Damage Detection'[repair_status] = "Temporary"
)
```

**7. FILL POTHOLE COUNT**

Counts incidents where potholes were immediately filled. Provides insight into responsiveness and minor maintenance efficiency.

```
Fill Pothole Count =
CALCULATE(
    COUNTROWS('Road Damage Detection'),
    'Road Damage Detection'[immediate_action] = "Fill Pothole"
)
```

**8. RESURFACE ROAD COUNT**

Tracks the number of incidents requiring complete road resurfacing. Highlights severe damage and supports large-scale maintenance planning.

```
Resurface Road Count =
CALCULATE(
    COUNTROWS('Road Damage Detection'),
    'Road Damage Detection'[immediate_action] = "Resurface Road"
)
```

## 9. IMMEDIATE BARRICADE COUNT

Counts incidents where urgent barricades were implemented. Measures preventive safety actions on high-risk roads.

```
Immediate Barricade Count =
CALCULATE(
   COUNTROWS('Road Damage Detection'),
   'Road Damage Detection'[immediate_action] = "Immediate Barricade"
)
```

## 10. AUTHORITY STAR RATING

Visualizes the performance of responsible authorities using a five-star system. Normalizes repair data to indicate efficiency and accountability.

```
Count of authority_responsible star rating =
VAR __MAX_NUMBER_OF_STARS = 5
VAR __MIN_RATED_VALUE = 0
VAR __MAX_RATED_VALUE = 100
VAR __BASE_VALUE = COUNTA('Road Damage Detection'[authority_responsible])
VAR __NORMALIZED_BASE_VALUE =
   MIN(
      MAX(
         DIVIDE(
            __BASE_VALUE - __MIN_RATED_VALUE,
            __MAX_RATED_VALUE - __MIN_RATED_VALUE
         ),
         0
```

```
        ),
        1
    )
VAR __STAR_RATING = ROUND(__NORMALIZED_BASE_VALUE * __MAX_NUMBER_OF_STARS, 0)
RETURN
    IF(
        NOT ISBLANK(__BASE_VALUE),
        REPT(UNICHAR(9733), __STAR_RATING)
            & REPT(UNICHAR(9734), __MAX_NUMBER_OF_STARS - __STAR_RATING)
    )
```

**11. AVERAGE RAINFALL**

Computes mean rainfall for all incidents. Supports correlation analysis between rainfall and road damage trends.

```
Avg Rainfall =
AVERAGE('Road Damage Detection'[rainfall_mm])
```

**12. HIGH RISK ROADS**

Counts all incidents categorized as high accident risk. Useful for identifying roads needing urgent safety interventions.

```
High Risk Roads =
CALCULATE(
    COUNT('Road Damage Detection'[id]),
    'Road Damage Detection'[accident_risk] = "High")
```

## 13. DAMAGE VS LAST MONTH

Compares current and previous month's total damage. Displays percentage change with arrows for quick trend visualization.

```
Damage vs Last Month =
VAR _CM = [Total Road Damage]
VAR _PM = [Previous Month Damage]
VAR _perc = DIVIDE(_CM - _PM, _PM)
VAR _format =
SWITCH(
   TRUE(),
   _perc > 0, UNICHAR(9650) & " " & FORMAT(_perc,"0.0%"),
   _perc < 0, UNICHAR(9660) & " " & FORMAT(_perc,"0.0%"),
   FORMAT(_perc,"0.0%")
)
RETURN
_format
```

## 14. CRITICAL RISK ROADS

Counts all road segments with critical accident risk. Helps identify dangerous areas requiring immediate attention. Supports prioritization of safety interventions.

```
Critical Risk Roads =
CALCULATE(
   COUNT('Road Damage Detection'[id]),
   'Road Damage Detection'[accident_risk] = "Critical")
```

## 15. AVERAGE RESPONSE TIME

Calculates the mean response time for repairing reported road damage. Evaluates maintenance efficiency and helps improve operational workflows.

```
Avg Response Time =
AVERAGE('Road Damage Detection'[response_time_hours])
```

## 16. TOTAL ACCIDENTS

Calculates the total number of accidents recorded in the dataset. Helps authorities track overall road safety trends and identify high-incident areas.

```
Total Accidents =
SUM('Road Damage Detection'[accident_history_count])
```

## 17. RISK SCORE

Computes a composite score by weighing potholes and cracks to quantify road hazard levels. Supports prioritization of maintenance and safety interventions.

```
Risk Score =
('Road Damage Detection'[Total Potholes] * 2) +
('Road Damage Detection'[Total Cracks] * 1)
```

# PUBLICATIONS

## Road Damage and Accident Risk Analytics Dashboard using Business Intelligence

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**Abstract:**

Road infrastructure monitoring and maintenance play a critical role in ensuring transportation safety and urban development. Inefficient repair tracking, unmanaged road damage, and delayed maintenance responses contribute significantly to accident risk and economic loss. This study presents a comprehensive visual analytics framework developed using Power BI to analyze road damage, accident patterns, repair status, and administrative-level infrastructure performance across municipalities and panchayat regions. The proposed interactive dashboard integrates multi-dimensional data attributes including traffic density, weather conditions, vehicle speed limits, accident history, repair status, and local governance units. Key analytical components include accident risk correlation, weather–traffic interaction matrices, vehicle speed risk indicators, geographical mapping of high-risk zones, and repair status monitoring across municipalities and rural panchayat regions. The system enables comparative analysis between repaired and pending road segments, identifies accident-prone administrative regions, and highlights maintenance gaps in urban and rural areas. Interactive slicers and conditional formatting enhance decision-making by allowing dynamic filtering across traffic level, weather conditions, and repair categories. The analysis reveals that accident frequency is significantly influenced by the combined impact of high traffic density, adverse weather conditions, and elevated speed limits. Furthermore, regions with delayed repair status exhibit higher risk concentration, emphasizing the importance of proactive infrastructure management. This study demonstrates how business intelligence tools can transform structured transportation datasets into actionable decision-support systems. The developed framework supports data-driven planning for municipalities and panchayat authorities, contributing to improved road safety, efficient maintenance allocation, and sustainable infrastructure governance.

**Keywords:**

Road Infrastructure Analytics – Power BI Dashboard – Business Intelligence System – Accident Risk Assessment – Road Damage Monitoring – Repair Status Tracking – Municipal Infrastructure Management – Panchayat-Level Governance Analytics – Traffic Density Analysis – Weather Impact Assessment – Vehicle Speed Risk Evaluation – Geospatial Risk Mapping – Interactive Data Visualization – Conditional Formatting Analysis – KPI-Based Monitoring – Data-Driven Decision Support – Transportation Safety Management – Urban and Rural Infrastructure Planning – Digital Infrastructure Governance – Smart City Analytics.

### I. INTRODUCTION

Road infrastructure is a fundamental component of socio-economic development, enabling mobility, trade, emergency response, and regional connectivity. The efficiency and safety of transportation systems depend heavily on the structural integrity and maintenance quality of road networks. However, rapid urban expansion, increasing vehicular density, environmental stressors, and administrative resource constraints have intensified the challenges associated with road monitoring and maintenance management. Inadequate infrastructure surveillance, delayed repair interventions, and fragmented reporting systems contribute significantly to accident risks, operational inefficiencies, and economic losses across both urban municipalities and rural panchayat regions.

Road damage, including surface cracks, potholes, and structural deterioration, directly impacts traffic flow and vehicle stability. Studies indicate that accident frequency often correlates with poor road conditions, adverse weather patterns, and excessive speed levels. Despite the availability of structured transportation

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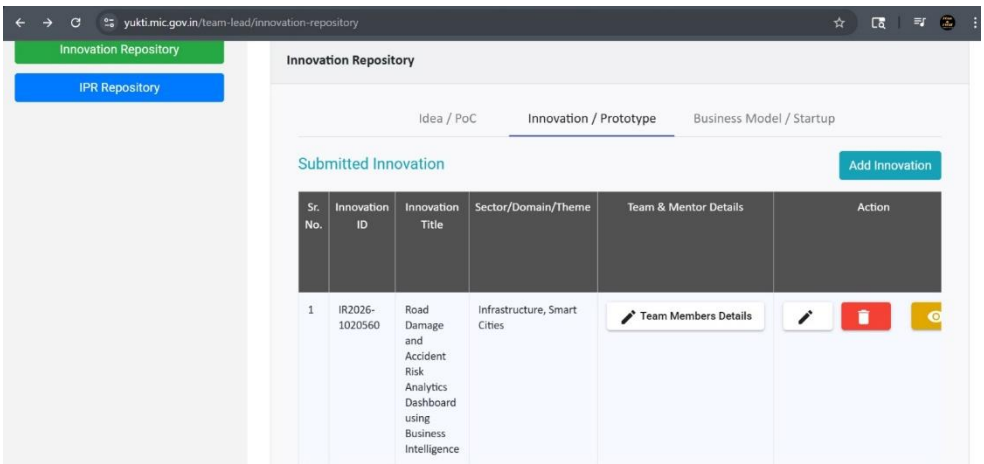

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